analysis.py > ...

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

df = pd.read_csv("dataset.csv")

print("\nFirst 5 Rows:")
print(df.head())

print("\nShape:")
print(df.shape)

print("\nColumns:")
print(df.columns.tolist())

print("\nData Types:")
print(df.dtypes)

print("\nDataset Information:")
df.info()

print("\nStatistical Summary:")
print(df.describe())

print("\nMissing Values:")
print(df.isnull().sum())

print("\nDuplicate Rows:")
print(df.duplicated().sum())

# Remove duplicates
df = df.drop_duplicates()

# Numerical columns
numeric_columns = df.select_dtypes(include=np.number).columns
```

```python
for column in numeric_columns:
    df[column] = df[column].fillna(df[column].mean())

# Categorical columns
categorical_columns = df.select_dtypes(include="object").columns

for column in categorical_columns:
    if df[column].isnull().sum() > 0:
        df[column] = df[column].fillna(df[column].mode()[0])

print("\nMissing Values After Cleaning:")
print(df.isnull().sum())

# Display categorical columns
print("\nCategorical Columns:")
print(df.select_dtypes(include="object").columns.tolist())

# Histogram
df.hist(figsize=(12, 10))
plt.tight_layout()
plt.show()

# Save cleaned dataset
df.to_csv("cleaned_dataset.csv", index=False)

print("\nCleaned dataset saved as cleaned_dataset.csv")
print("Final Shape:", df.shape)
```

```
PS C:\Users\srush\OneDrive\Desktop\ML Project> python analysis.py

First 5 Rows:
    Destination Name          State Region   Category Popular Attraction Accessibility  ... Adventure Historical  Beach  Wildlife  Family_Friendly  Nightlife
0          Taj Mahal  Uttar Pradesh  North   Heritage          Taj Mahal          Easy  ...         1          5      1         1                5          1
1             Jaipur      Rajasthan   West   Heritage         Amber Fort          Easy  ...         1          5      1         1                5          1
2                Goa            Goa   West      Beach     Calangute Beach          Easy  ...         4          1      5         1                4          4
3  Kerala Backwaters         Kerala  South     Nature      Vembanad Lake      Moderate  ...         3          1      1         1                4          1
4           Varanasi  Uttar Pradesh  North  Religious  Ghats of Varanasi      Moderate  ...         1          4      1         1                5          1

[5 rows x 15 columns]

Shape:
(110, 15)

Columns:
['Destination Name', 'State', 'Region', 'Category', 'Popular Attraction', 'Accessibility', 'Nearest Airport', 'Nearest Railway Station', 'Nature', 'Adventure', 'Historical', 'Beach', '
Wildlife', 'Family_Friendly', 'Nightlife']

Data Types:
Destination Name            str
State                       str
Region                      str
Category                    str
Popular Attraction          str
Accessibility               str
Nearest Airport             str
Nearest Railway Station     str
Nature                    int64
Adventure                 int64
Historical                int64
Beach                     int64
Wildlife                  int64
Family_Friendly           int64
Nightlife                 int64
dtype: object
```

```
Dataset Information:
<class 'pandas.DataFrame'>
RangeIndex: 110 entries, 0 to 109
Data columns (total 15 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Destination Name        110 non-null    str
 1   State                   110 non-null    str
 2   Region                  110 non-null    str
 3   Category                110 non-null    str
 4   Popular Attraction      110 non-null    str
 5   Accessibility           110 non-null    str
 6   Nearest Airport         110 non-null    str
 7   Nearest Railway Station  110 non-null    str
 8   Nature                  110 non-null    int64
 9   Adventure               110 non-null    int64
 10  Historical              110 non-null    int64
 11  Beach                   110 non-null    int64
 12  Wildlife                110 non-null    int64
 13  Family_Friendly         110 non-null    int64
 14  Nightlife               110 non-null    int64
dtypes: int64(7), str(8)
memory usage: 13.0 KB

Statistical Summary:
          Nature   Adventure  Historical       Beach    Wildlife  Family_Friendly   Nightlife
count  110.000000  110.000000  110.000000  110.000000  110.000000       110.000000  110.000000
mean     3.200000    2.245455    2.645455    1.618182    1.590909         3.990909    1.618182
std      1.584327    1.293501    1.588874    1.306173    1.127598         0.840440    1.031449
min      1.000000    1.000000    1.000000    1.000000    1.000000         1.000000    1.000000
25%      2.000000    1.000000    1.000000    1.000000    1.000000         4.000000    1.000000
50%      3.500000    2.000000    2.000000    1.000000    1.000000         4.000000    1.000000
75%      5.000000    3.000000    4.000000    1.000000    2.000000         5.000000    2.000000
max      5.000000    5.000000    5.000000    5.000000    5.000000         5.000000    5.000000
```

```
Missing Values:
Destination Name          0
State                     0
Region                    0
Category                  0
Popular Attraction        0
Accessibility             0
Nearest Airport           0
Nearest Railway Station   0
Nature                    0
Adventure                 0
Historical                0
Beach                     0
Wildlife                  0
Family_Friendly           0
Nightlife                 0
dtype: int64

Duplicate Rows:
0
C:\Users\srush\OneDrive\Desktop\ML Project\analysis.py:41: Pandas4Warning: For backward compatibility, 'str' dtypes are included by select_dtypes when 'object' dtype is spe
cified. This behavior is deprecated and will be removed in a future version. Explicitly pass 'str' to `include` to select them, or to `exclude` to remove them and silence t
his warning.
See https://pandas.pydata.org/docs/user_guide/migration-3-strings.html#string-migration-select-dtypes for details on how to write code that works with pandas 2 and 3.
  categorical_columns = df.select_dtypes(include="object").columns
```

```
Missing Values After Cleaning:
Destination Name           0
State                      0
Region                     0
Category                   0
Popular Attraction         0
Accessibility              0
Nearest Airport            0
Nearest Railway Station    0
Nature                     0
Adventure                  0
Historical                 0
Beach                      0
Wildlife                   0
Family_Friendly            0
Nightlife                  0
dtype: int64

Categorical Columns:
C:\Users\srush\OneDrive\Desktop\ML Project\analysis.py:52: Pandas4Warning: For backward compatibility, 'str' dtypes are included by select_dtypes when 'object' dtype is spe
cified. This behavior is deprecated and will be removed in a future version. Explicitly pass 'str' to `include` to select them, or to `exclude` to remove them and silence t
his warning.
See https://pandas.pydata.org/docs/user_guide/migration-3-strings.html#string-migration-select-dtypes for details on how to write code that works with pandas 2 and 3.
  print(df.select_dtypes(include="object").columns.tolist())
['Destination Name', 'State', 'Region', 'Category', 'Popular Attraction', 'Accessibility', 'Nearest Airport', 'Nearest Railway Station']

Cleaned dataset saved as cleaned_dataset.csv
Final Shape: (110, 15)
```

